



B-Tech- II

Department of Science and Humanities



CLASS -2

UNIT 3

Laplace transform



Laplace Transform of some Elementary Functions

$$(1) L(1) = \frac{1}{s}$$

Proof : -By Definition

$$L(1) = \int_0^{\infty} e^{-st} \cdot 1 dt = \left[\frac{e^{-st}}{-s} \right]_0^{\infty} = \frac{1}{s}, (s > 0)$$

$$(2) L(e^{at}) = \frac{1}{s-a}$$

Proof : -By Definition

$$\begin{aligned} L(e^{at}) &= \int_0^{\infty} e^{-st} \cdot e^{at} dt = \int_0^{\infty} e^{-(s-a)t} dt \\ &= \left[\frac{e^{-(s-a)t}}{-(s-a)} \right]_0^{\infty} = \frac{1}{s-a} \text{ if } s > a \end{aligned}$$

$$(3) L[e^{-at}] = \frac{1}{s+a}, s > -a$$

$$(4) L[\sinh at] = \frac{a}{s^2 - a^2}$$

Proof : - We have $\sinh at = \frac{e^{at} - e^{-at}}{2}$ and $\cosh at = \frac{e^{at} + e^{-at}}{2}$

By definition

$$\begin{aligned} L(\sinh at) &= L\left(\frac{e^{at} - e^{-at}}{2}\right) = \frac{1}{2}[L(e^{at}) - L(e^{-at})] \\ &= \frac{1}{2}\left[\frac{1}{s-a} - \frac{1}{s+a}\right] \\ &= \frac{a}{s^2 - a^2}, s > |a| \end{aligned}$$

$$(5) \text{ Similarly, } L[\cosh at] = \frac{s}{s^2 - a^2}, s > |a|$$

$$(6) L[\sin at] = \frac{a}{s^2 + a^2} \text{ and } L[\cos at] = \frac{s}{s^2 + a^2}, s > 0$$

Proof: - We know that $e^{ix} = \cos x + i \sin x$ [Euler's Formula]

$$\therefore e^{iat} = \cos at + i \sin at$$

$$\begin{aligned} \therefore L[\cos at + i \sin at] &= L[e^{iat}] = \frac{1}{s - ia} \left[\Theta L(e^{at}) = \frac{1}{s - a} \right] \\ &= \frac{s + ia}{s^2 + a^2} = \frac{s}{s^2 + a^2} + i \frac{a}{s^2 + a^2} \end{aligned}$$

Equating real and imaginary parts, we get

$$L[\sin at] = \frac{a}{s^2 + a^2} \text{ and } (7) L[\cos at] = \frac{s}{s^2 + a^2}, s > 0$$

$$(8) L(t^n) = \frac{n!}{s^{n+1}} \text{ or } \frac{n!}{s^{n+1}}$$

$$\text{Proof: } -L(t^n) = \int_0^{\infty} e^{-st} t^n dt, \text{ putting } st = u$$

$$= \int_0^{\infty} e^{-u} \left(\frac{u}{s}\right)^n \frac{du}{s}$$

$$= \frac{1}{s^{n+1}} \int_0^{\infty} e^{-u} u^{(n+1)-1} du$$

$$\therefore L(t^n) = \frac{n!}{s^{n+1}}, n > -1 \quad \left[\Theta \overline{n} = \int_0^{\infty} e^{-x} x^{n-1} dx, n > 0 \right]$$

$$\text{or } L(t^n) = \frac{n!}{s^{n+1}} \quad [n = 0, 1, 2, \dots] \quad [\Theta n+1 = n!]$$

Definition and results of Gamma function

- Gamma function $\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx$

$\Gamma(n+1) = n!$ when n is a positive integer

- Hence $L[t^n] = \frac{n!}{s^{n+1}}$

$\Gamma(n+1) = n\Gamma(n)$ when n is a positive or negative fraction

- $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$



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Laplace transform of some elementary functions

$$L\{1\} = \frac{1}{s}$$

$$L\{t^n\} = \frac{n!}{s^{n+1}}; n \in \mathbb{N}$$

$$L\{e^{at}\} = \frac{1}{s-a}; s > a$$

$$L\{t^n\} = \Gamma \frac{(n+1)}{s^{n+1}}; n > -1$$

$$L\{\cos at\} = \frac{s}{s^2 + a^2}$$

$$L\{\cosh at\} = \frac{s}{s^2 - a^2}$$

$$L\{\sin at\} = \frac{a}{s^2 + a^2}$$

$$L\{\sinh at\} = \frac{a}{s^2 - a^2}$$



Example:

Find the Laplace transform of
 $f(t) = 5e^{-2t} - 3\sin(4t)$ for $t \geq 0$.

Solution:

$$\begin{aligned} F(s) &= L\{f(t)\} \\ &= L\{5e^{-2t} - 3\sin(4t)\} \\ &= 5L\{e^{-2t}\} - 3L\{\sin(4t)\} \\ &= \frac{5}{s+2} - \frac{12}{s^2+16}, \quad s > 0 \end{aligned}$$





THANK YOU

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